A Diagnostic Tool for ATTR using Bone Scintigraphy Scans

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Predicting

- Whole-body bone scintigraphy can reveal diffuse cardiac uptake in ATTR-CM and focal or diffuse skeletal uptake in metastatic disease, but the visual patterns can still be subtle.
- We built a three-way classifier for control vs. ATTR-CM vs. metastatic bone cancer on 1,000 synthetic scans.
- The project compares an interpretable radiomics-lite logistic regression baseline, a custom convolutional neural network, and a transfer-learning DINOv2 ViT-S/14 baseline.
- All three models performed well, and the pretrained ViT achieved the strongest held-out test performance.
- We also generated attribution visualizations so the model decisions could be tied back to uptake patterns in the scan.

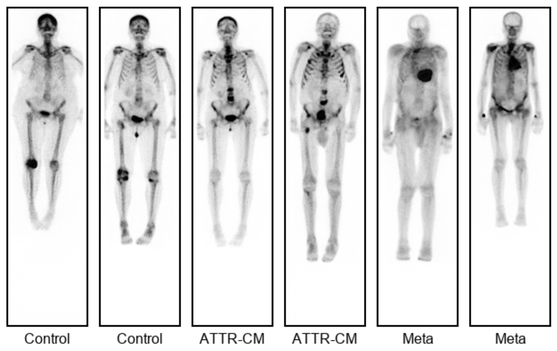
Data

- The data come from an open synthetic whole-body bone scintigraphy release on Zenodo [1]; locally, the dataset consists of 1,000 grayscale PNG scans plus a `key_label.csv` file that maps each image key to a ground-truth label.
- We use a stratified split of 698 training, 151 validation, and 151 test images across control, ATTR-CM, and metastatic bone cancer.
- Each example is a single anterior-projection scan at native resolution 1024×256, so there is no time-series discretization step.
- For preprocessing, logistic regression uses intensities scaled to $[0, 1]$, the CNN resizes scans to 512×128 and applies train-only normalization and mild augmentation, and the ViT replicates grayscale into RGB, pads scans to 336×336, and uses pretrained DINOv2 normalization.

Features

- The logistic-regression baseline uses 51 radiomics-lite features: global intensity statistics, approximate heart/rib/spine/skull/pelvis ROI summaries, and simple derived ratios such as heart-to-rib uptake and upper-to-lower uptake.
- These features are appropriate because ATTR-CM often changes cardiac uptake while metastatic disease changes the spatial distribution and intensity of osseous hotspots.
- The CNN and ViT instead use raw image pixels as input and learn features directly from the scan, so we can compare handcrafted structure against learned spatial representations.
- We do not use HOG, PCA, ICA, Fourier features, or other generic feature transforms in this project.

Dataset Examples



One representative scan from each class.

Models

- All three models output logits $z \in \mathbb{R}^3$ and use the same softmax classifier:

$$p(y = k \mid x; \theta) = \frac{\exp(z_k)}{\sum_{j=0}^{2} \exp(z_j)}, \qquad \mathcal{L}(\theta) = -\frac{1}{n} \sum_{i=1}^{n} \alpha_{y_i} \log p(y_i \mid x_i; \theta),$$

Radiomics Logistic Regression

- The radiomics baseline uses a 51-dimensional handcrafted feature vector and a linear multiclass classifier:

$$z = Wx + b.$$

Convolutional Neural Network

- The CNN takes resized grayscale scans and applies four convolutional blocks with channel sizes $32 \to 64 \to 128 \to 256$, followed by adaptive average pooling and a small MLP head.

$$h^{(\ell)} = \mathrm{Pool}\Big(\mathrm{ReLU}\Big(\mathrm{BN}\Big(W^{(\ell)} * h^{(\ell-1)} + b^{(\ell)}\Big)\Big)\Big),$$

DINOv2 ViT-S/14

- The ViT baseline uses a pretrained DINOv2-small backbone with a new three-class linear head, while the backbone is frozen in the first-pass transfer-learning baseline.
- Its core self-attention operation is

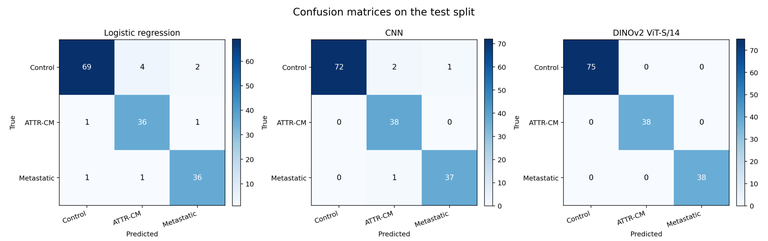
$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

- In practice, grayscale scans are replicated into three channels and padded to a square input before being passed through the pretrained model.

Results

- Table 1 reports final train and held-out test error; the train and test sets contain 698 and 151 scans, respectively.
- The same ranking held across the richer test metrics: DINOv2 was best, followed by the CNN, then radiomics logistic regression.

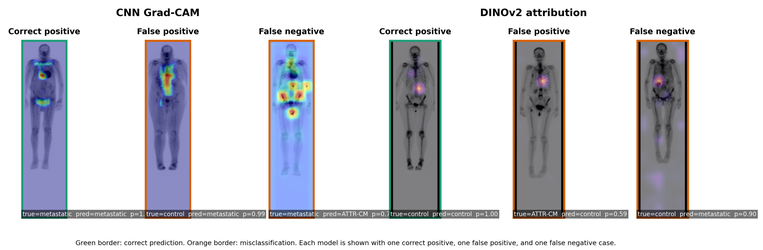
Model	Train Err	Test Err	Test Acc	Macro-F1	AUROC	Log Loss
Logistic regression	4.44%	6.62%	0.934	0.931	0.994	0.152
CNN	4.73%	2.65%	0.974	0.972	0.996	0.095
DINOv2 ViT-S/14	0.72%	0.00%	1.000	1.000	1.000	0.029



Confusion matrices on the held-out test split. The ViT removes all off-diagonal errors.

Qualitative Results

- The figure below compares matched qualitative cases for the CNN and DINOv2 models.
- Each model is shown as a left-to-right sequence of three cases: correct positive, false positive, and false negative.
- In the correct positive cases, both models place their strongest emphasis on clinically meaningful uptake regions rather than diffuse background structure.
- In the false positive and false negative cases, the highlighted regions become more diffuse or shift toward ambiguous uptake patterns, which helps explain the errors.



Green border: correct prediction. Orange border: misclassification. Each model is shown with one correct positive, one false positive, and one false negative case.

Left: CNN Grad-CAM. Right: DINOv2 patch attribution.

Discussion

- We expected the learned image models to outperform the handcrafted baseline, and that is what we observed.
- Logistic regression already worked well because the synthetic classes differ in interpretable uptake statistics, especially in the heart and skeleton.
- The CNN improved on that baseline by learning nonlinear spatial structure directly from the image instead of relying on fixed ROIs.
- The frozen DINOv2 model performed best, which suggests that transfer learning and global patch interactions are very effective even for grayscale medical scans after simple channel replication.
- Because the dataset is synthetic and relatively clean, the perfect DINOv2 test score should be interpreted cautiously rather than as proof of clinical readiness.
- We mitigated overfitting with train-only normalization, data augmentation, weight decay, and validation-based early stopping.

Future Work

- With six more months, we would first test the models on a harder external dataset or a synthetic-to-real transfer setting.
- We would also partially unfreeze the last DINOv2 transformer blocks, run broader hyperparameter tuning or cross-validation, and compare additional medical-imaging backbones.
- A final priority would be deeper interpretability, calibration, and robustness analysis for borderline cases.

References

[1] David Haberl et al.
Generative artificial intelligence enables the generation of bone scintigraphy images and improves generalization of deep learning models in data-constrained environments.
52(7):2355–2368, 2025.

[2] Rohit Mohanan Kumbalath et al.
Artificial intelligence in cardiac amyloidosis: A systematic review and meta-analysis of diagnostic accuracy across imaging and non-imaging modalities.
medRxiv, 2025.